Pabbuleti Venkata Charan Teja

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.54
Sri Chaitanya Junior College	Class XII (State (BIEAP))	2024	97.10%
Sri Chaitanya School	Class X (State (BSEAP))	2022	95.50%

RELEVANT COURSES

Mathematics: Matrix Theory, Probability and Random Variables, Optimization I & II, Numerical Analysis

Core AI: Foundations of Machine Learning, Deep Learning, AI for Humanity

Computer Science: Discrete Mathematics, Data Structures & Applications, Algorithms, Database Management Systems

Systems and Engineering: Linear Systems and Signal Processing, Control Systems

PROJECTS

MiniML: A NumPy-Based Machine Learning Framework | *GitHub Link*

- Architected a modular machine learning framework entirely from **NumPy**, implementing vectorized optimization algorithms, lightweight automatic differentiation, reusable training pipelines, and extensible machine learning utilities.
- Developed reusable neural network primitives including **Sequential containers, parameterized linear layers, ReLU and Sigmoid activations, custom backpropagation, configurable optimizers, and modular training workflows** without external ML libraries.
- Engineered a comprehensive ML toolkit featuring **Ridge Regression, Logistic Regression, Gaussian Naive Bayes, PCA, polynomial and Gaussian feature transformations, preprocessing utilities, model selection, and K-Fold cross-validation**.
- Evaluated the framework on **Fashion-MNIST**, validating implementation correctness against established machine learning baselines while ensuring stable convergence, reproducible experimentation, and reliable optimization performance.

Deep Learning From Scratch: A NumPy-Based Deep Learning Framework | *GitHub Link*

- Architected a modular deep learning framework entirely from **NumPy**, implementing reusable computational layers, configurable training pipelines, and complete forward and backward propagation without high-level deep learning libraries.
- Developed custom implementations of **MLPs, CNNs, Vanilla RNNs, LSTMs, and Coupled LSTMs**, including Backpropagation Through Time (BPTT), modular network construction, and reusable sequence learning components.
- Engineered reusable deep learning modules including **Dense, Conv2D, Pooling, Flatten, Sequential containers, SGD and Adam optimizers, Xavier and He initialization, Cross-Entropy losses, L1/L2 regularization, gradient clipping, and early stopping**.
- Evaluated models on **MNIST** and **IMDB**, achieving **98.24%** test accuracy with CNN on MNIST and **85.52%** test accuracy with Coupled LSTM on IMDB while validating recurrent implementations on synthetic sequence learning benchmarks.

Synthetic Shape Classification: A Multi-Scale Deep Learning Pipeline | *GitHub Link*

- Developed a modular computer vision pipeline in **PyTorch**, implementing a configurable residual CNN with progressive feature extraction, Global Average Pooling, residual connections, and reusable training components.
- Engineered an end-to-end training workflow featuring **multi-scale learning, stratified 5-fold cross-validation, test-time augmentation, threshold optimization, confidence-based prediction calibration, automated checkpointing, and comprehensive evaluation**.

- Designed a reusable software architecture with modular datasets, preprocessing pipelines, configurable training engine, experiment management, evaluation utilities, reproducible workflows, and comprehensive project documentation.

SKILLS

Programming Languages: Python, C

AI & ML: Machine Learning, Deep Learning, Active Learning, Computer Vision, Fairness in Machine Learning

Databases: PostgreSQL

Web Technologies: HTML, CSS

Tools And Platforms: Git, GitHub, Linux, LaTeX, VS Code, Jupyter Notebook, pgAdmin

POSITIONS OF RESPONSIBILITY

Inspection Coordinator | Mess Monitoring Council, IIT Hyderabad

- Led periodic mess inspections to ensure hygiene standards, food quality, operational compliance, prompt resolution of maintenance issues, and continuous improvement of overall dining services.
- Coordinated with mess staff, student representatives, and administration to streamline service delivery, resolve operational issues, and address student concerns through timely corrective actions.